# MEMANTINE

Moderate–severe Alzheimer's
Renal dose adjustment

**CHEMICAL NAME** 3,5-dimethyladamantan-1-amine (hydrochloride)
**CLASS** NMDA receptor antagonist (anti-dementia)
**TRADE NAMES** Ebixa · Namenda · Axura · Marixino

## THERAPEUTIC USES / INDICATIONS

- Moderate to severe Alzheimer's disease (alone or with an acetylcholinesterase inhibitor).

## PHARMACODYNAMICS

- Uncompetitive **NMDA receptor antagonist** → blocks pathological glutamate excitotoxicity while allowing normal signalling.

## ADMINISTRATION

- **PO** once daily, with or without food; oral solution available.
- Weekly titration pack.

## DOSAGES

- **Titration:** 5 mg daily week 1, 10 mg week 2, 15 mg week 3, 20 mg week 4 onward.
- **CrCl 30–49:** 10 mg daily (may increase to 20 mg if tolerated); **CrCl 5–29:** max 10 mg.

## PHARMACOKINETICS

- **Peak:** 3–8 h. Bioavailability ~100%.
- Minimal metabolism.
- Excreted unchanged in urine (↓ with alkaline urine). **t½:** 60–100 h.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Dizziness, headache.
- Constipation, somnolence.
- Hypertension.

### ADVERSE EFFECTS (SERIOUS)

- Confusion, hallucinations, agitation.
- Seizures (rare).
- Falls.

## CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** epilepsy, severe renal impairment, conditions raising urine pH.

### KEY INTERACTIONS

- **Amantadine, ketamine, dextromethorphan:** additive NMDA effects — avoid.
- **Urine alkalinisers (bicarbonate):** ↑ levels.
- **Warfarin:** monitor INR.

## NURSING MANAGEMENT

- Follow titration; check renal function.
- Monitor BP, dizziness, falls, behaviour.
- Assess benefit at review.

## PATIENT EDUCATION

- Increase dose weekly as instructed.
- Report dizziness, confusion or seizures.
- Carers: supervise daily dosing.

**KEY POINTS**

- NMDA antagonist; moderate–severe Alzheimer's.
- Renal dose adjustment.
- Titrate 5 mg weekly to 20 mg.
- Dizziness, constipation, confusion.

**SOURCES CHECKED** **[830]** Memantine 20 mg film-coated tablets (UK SmPC) **[831]** Memantine Hydrochloride, oral (US label) **[832]** memantine hydrochloride, oral (US label)
*Full references p. 432–457*